ON THE DIFFEOLOGY OF ORBIT SPACES

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Dedicated to Richard S. Palais on the occasion of his 94th birthday.
This work is presented in deep appreciation of his foundational contributions
to the theory of transformation groups, which form the essential framework of this paper.

ABSTRACT. We investigate the correspondence between the geometry of a smooth action of a compact Lie group on a manifold M and the intrinsic smooth structure of the orbit space M/G. The latter is captured by the *Klein stratification*, which partitions M/G into orbits of local diffeomorphisms, classifying the space by its intrinsic singularity types. We show that this geometry maps coherently to the quotient through the *isostabilizer decomposition* of M, whose elements are the connected components of submanifolds where the stabilizer is constant. Our main result, the *Correspondence Theorem*, establishes that the canonical projection induces a surjective map from this decomposition to the Klein strata. As a corollary, we define the *Inverse Klein Stratification* on M by pullback, clarifying its relationship with the intrinsic geometry of the quotient.

INTRODUCTION

The study of spaces arising from the smooth action of a compact Lie group G on a manifold M is a cornerstone of differential geometry, with a rich history built upon the work of Palais, Bredon, and Mostov. A central challenge is that the orbit space M/G is generally a singular space. Classically, the action on M is organized through stabilizers and their slice representations, as described by the Slice Theorem [Bre72, Pal60]. This local structure provides a precise description of neighborhoods of orbits in terms of associated bundles $G \times_K E$, where $K = \mathrm{Stab}_G(x)$ acts linearly on the normal slice E. Passing to the quotient, this yields local models of the form E/K for M/G.

However, while these constructions describe the geometry of M and its quotient from the perspective of the group action, they do not directly address the intrinsic smooth structure of the quotient space itself. They answer the question of how the quotient is built from M, but not necessarily what they "look like" locally.

Date: May 8, 2026.

2020 *Mathematics Subject Classification.* 57S15, 58A40, 58A35.

Key words and phrases. Diffeology, Orbit Space, Isostabilizer Decomposition, Klein Stratification, Orthofold.

P.I-Z thanks the Hebrew University of Jerusalem, Israel, for its continuous academic support. He would also like to acknowledge the assistance provided by the AI assistants Gemini (Google) and DeepSeek (深度求索).

The framework of diffeology offers a different perspective. It provides a powerful, intrinsic theory of smoothness for the quotient space M/G itself. This intrinsic geometry is captured by the *Klein stratification*, the partition of M/G into its local diffeomorphism orbits, which classifies the space by its intrinsic singularity types. This raises a fundamental question: what canonical partition of M maps coherently to the Klein stratification of its quotient?

One might hope that grouping points merely by their stabilizer subgroups would suffice, but our key examples show that this is not the case. The local geometry of M/G depends not only on the stabilizer subgroup $\mathrm{Stab}_G(x)$ but also on its slice representation. At the same time, points with different stabilizers may nevertheless produce locally diffeomorphic quotient models. Thus, neither the stabilizer subgroup alone nor its conjugacy class fully captures the intrinsic geometry of the quotient.

We show in this paper that to map the smooth structure of M coherently to the quotient via the projection orbit : $M \to M/G$, the natural tool is the finer *isostabilizer decomposition*, whose elements are the connected components of submanifolds where the stabilizer subgroup is constant. On each such component, the slice representation is fixed up to equivalence, making these submanifolds the fundamental smooth "atoms" of the action from the perspective of quotient geometry.

We establish the main result of this paper: the *Correspondence Theorem*, which states that the canonical projection induces a surjective map from the isostabilizer decomposition of M to the Klein stratification of M/G. This provides the systematic dictionary between the local data of the action — the stabilizer and its slice representation — and the intrinsic singularity types of the quotient.

As a consequence of our Correspondence Theorem, we are able to define a new, canonical stratification on M itself—the *Inverse Klein Stratification*—by pulling back the Klein strata from the quotient. This allows us to clarify the hierarchy of partitions on M: the isostabilizer decomposition is the finest, refining the Inverse Klein Stratification, while the latter reflects exactly the intrinsic geometry retained by the quotient. Our analysis also highlights the feature that the intrinsic smooth identity of M/G, which we term an *orthofold*, can be independent of the specific global action that created it.

As we will see, the intrinsic geometry of the quotient M/G is assembled naturally according to three kinds of points, characterized by their local diffeological models:

- **Manifold points**, whose local model is a Euclidean space E (i.e., the effective stabilizer is trivial). These correspond to principal orbits.
- **Orbifold points**, whose local model is E/Γ where Γ is a non-trivial finite group. These correspond to exceptional orbits.
- **Orthofold points**, whose local model is E/H where H is a non-discrete compact Lie group. These correspond to non-exceptional singular orbits.

This classification provides a hierarchy of singularities that is intrinsic to the quotient and independent of the original action.

The paper is organized as follows: Section I provides preliminaries on diffeology. Section II discusses the diffeological structure of M/G. Section III presents the illustrative examples that motivate our analysis. Section IV defines the isostabilizer decomposition. Section V establishes our main theorem and, as a corollary, constructs the Inverse Klein Stratification. Finally, Section VI provides concluding remarks.

I. Preliminaries on Diffeology

In this section, we provide a brief introduction to the fundamental concepts of diffeology necessary for understanding the structure of orbit spaces as diffeological spaces. For a comprehensive treatment, we refer the reader to [PIZ13].

1. Diffeological Spaces.

A *parametrization* in a set X is a map $\mathrm{P} : \mathrm{U} \to \mathrm{X}$, where U is an open subset of some Euclidean space $\mathbf{R}^k$. A *diffeology* on a set X is a collection of parametrizations, called the *plots of the diffeology*, satisfying three axioms:

(1) **Covering:** Any constant map $\mathrm{P} : \mathrm{U} \to \mathrm{X}$ is a plot.
(2) **Locality:** If $\mathrm{P} : \mathrm{U} \to \mathrm{X}$ is a parametrization such that for every $r \in \mathrm{U}$, there exists an open neighborhood $\mathrm{V} \subseteq \mathrm{U}$ of r such that the restriction $\mathrm{P}|_\mathrm{V}$ is a plot, then P is a plot.
(3) **Smooth Compatibility:** If $\mathrm{P} : \mathrm{U} \to \mathrm{X}$ is a plot and $f : \mathrm{V} \to \mathrm{U}$ is a smooth map between open subsets of Euclidean spaces, then the composition $\mathrm{P} \circ f : \mathrm{V} \to \mathrm{X}$ is a plot.

A *diffeological space* is a set X equipped with a diffeology. Euclidean domains and smooth manifolds are first examples of diffeological spaces, where the plots are precisely the smooth maps (in the usual sense) from open subsets of Euclidean spaces into Euclidean spaces or manifold.

A map $f : \mathrm{X} \to \mathrm{Y}$ between diffeological spaces is *smooth* if for every plot P of X, the composition $f \circ \mathrm{P}$ is a plot of Y. A smooth map f is a *diffeomorphism* if it is bijective and its inverse f^{-1} is also smooth.

2. D-Topology.

Every diffeological space X carries a natural topology, called the *D-topology*. A subset $\mathcal{O} \subseteq \mathrm{X}$ is defined to be *D-open* if and only if for every plot $\mathrm{P} : \mathrm{U} \to \mathrm{X}$, the preimage $\mathrm{P}^{-1}(\mathcal{O})$ is an open subset of U in the standard Euclidean topology.

3. Subspaces and Quotients.

Given a diffeological space X and a subset $\mathrm{A} \subseteq \mathrm{X}$, the *subset diffeology* on A is defined by taking as plots all parametrizations $\mathrm{P} : \mathrm{U} \to \mathrm{A}$ such that P, viewed as a parametrization into X, is a plot of X.

Given a diffeological space X and an equivalence relation $\sim$ on X, the *quotient diffeology* on the set of equivalence classes $\mathrm{X}/\sim$ is defined as follows: a parametrization $\bar{\mathrm{P}} : \mathrm{U} \to \mathrm{X}/\sim$ is a plot if for every $r \in \mathrm{U}$, there exists an open neighborhood $\mathrm{V} \subseteq \mathrm{U}$ of r and a plot $\mathrm{P} : \mathrm{V} \to \mathrm{X}$ such that $\bar{\mathrm{P}}|_\mathrm{V}$ is the composition of P with the canonical projection $\pi : \mathrm{X} \to \mathrm{X}/\sim$.

The canonical projection $\pi : X \to X/\!\sim$ is a fundamental example of a *subduction*. A map $f : X \to Y$ between diffeological spaces is a subduction if it is smooth, surjective, and a parametrization $P : U \to Y$ is a plot of Y if and only if for every $r \in U$, there exists an open neighborhood $V \subseteq U$ of r and a plot $\tilde{P} : V \to X$ such that $P|_V = f \circ \tilde{P}$.

4. Local Diffeomorphisms and Invariants.

A *local diffeomorphism* from X to Y is a map f between two D-open subsets $U \subseteq X$ and $V \subseteq Y$ which is a diffeomorphism from U (with the subset diffeology) to V (with the subset diffeology). The collection of all local diffeomorphisms of a space X forms a pseudogroup, denoted $\mathrm{Diff}_{\mathrm{loc}}(X)$.

The diffeology of a space X determines several intrinsic invariants, in particular:

Definition (Dimension Map). *The* diffeological dimension *at a point $x \in X$, denoted* $\dim_x(X)$*, is an intrinsic invariant that measures the minimal dimension of a Euclidean space from which one can locally generate all plots around x. It is defined precisely as a minimax over families of plots pointed at x.*

This definition was introduced in [PIZ07] and is further elaborated with many examples in [PIZ13]. (See [PIZ13, Chapter 2] for the complete definition and properties.) It coincides with the standard dimension for manifolds.

Definition (Stratification). *A* stratification *of a diffeological space* X *is a partition into disjoint subsets (strata) satisfying the* Frontier Condition *with respect to the D-topology. This is the minimal requirement for a partition to be called a stratification; we therefore call it a* basic stratification *and label it with the condition:*

[B] *The closure of every stratum $S \in \mathscr{S}$ is a union of strata.*

Diffeological stratifications may satisfy other specific conditions, often represented by various labels (e.g., [LF] for locally finite, [M] for manifold strata). For a presentation of the general theory of stratifications in diffeology, we refer to [GIZ23].

In the spirit of Klein's Erlangen Program, the local geometry of a diffeological space is classified by its pseudogroup of local diffeomorphisms, $\mathrm{Diff}_{\mathrm{loc}}(X)$. The orbits of this pseudogroup partition the space X into sets of points that are locally diffeomorphic to each other. These orbits are called the *Klein strata* of X [PIZ13, §1.42]. We denote by $K\ell(X)$ the set of Klein strata of X. They form the finest partition of X that respects its local smooth structure.

Proposition (Klein Stratification is Basic). *The partition of a diffeological space into Klein strata is a basic stratification: The closure of a Klein stratum is a union of Klein strata.*

Proof. This proposition is a standard result in diffeology (see [PIZ13, §1.42] and [PIZ25, Section 74]). The proof relies on the fact that local diffeomorphisms are also homeomorphisms with respect to the D-topology. Since the Klein strata are the orbits of this pseudogroup, the closure of a stratum is necessarily a union of strata, thus satisfying the frontier condition. □

Points in the same Klein stratum share the same local geometry, understood as local diffeological structure. We can also say that the Klein stratification partitions the space into points with the same types of singularities. In particular,

Proposition (Klein Stratification and Dimension). *The diffeological dimension is constant on the Klein strata.*

Proof. By definition, any two points in the same Klein stratum are locally diffeomorphic. Since the diffeological dimension is an invariant under local diffeomorphisms (see [PIZ13, §2.22]), it must be constant on each Klein stratum. □

II. Diffeological Structure of Orbit Spaces and Dimension

In this section, we clarify the local structure of the diffeological quotient of a connected manifold M by the smooth action of a compact Lie group G. We then introduce the (intrinsic) diffeological dimension map on the orbit space M/G and establish a fundamental formula relating it to the dimensions of the manifold and its orbits.

5. Local Structure of Orbit Spaces.

Let G be a compact Lie group acting smoothly on a connected manifold M. Let $\mathscr{O} = \mathrm{G}\cdot x$ be the orbit of a point $x \in \mathrm{M}$, and let $\mathrm{H} = \mathrm{Stab}_{\mathrm{G}}(x)$ be its stabilizer.

The classical Slice Theorem (see, e.g., [Pal60, Bre72]) states that there exists a G-invariant neighborhood $\mathrm{V} \subset \mathrm{M}$ of the orbit $\mathscr{O}$, and a vector space E (the normal slice at x) on which H acts orthogonally, such that V is G-equivariantly diffeomorphic to the associated bundle $\mathrm{G} \times_{\mathrm{H}} \mathrm{E} = (\mathrm{G} \times \mathrm{E})/\mathrm{H}$.

When we equip M/G with the quotient diffeology, this local structure translates directly into a local diffeomorphic equivalence:

Proposition (Diffeological Local Model). *As a diffeological space, the quotient space* M/G *is locally diffeomorphic to some quotient* E/H *where* E *is a Euclidean space and* H *is a compact subgroup of* $\mathrm{O}(\mathrm{E})$ *acting orthogonally. Precisely, if* $\mathrm{V} \subset \mathrm{M}$ *is a* G*-invariant open neighborhood of an orbit* $\mathscr{O} = \mathrm{G}\cdot x$ *and* V *is* G*-equivariantly diffeomorphic to* $\mathrm{G} \times_{\mathrm{H}} \mathrm{E}$ *(where* $\mathrm{H} = \mathrm{Stab}_{\mathrm{G}}(x)$*), then the quotient space* V/G *(with the subset diffeology from* M/G*) is diffeomorphic to* E/H *(with the quotient diffeology).*

Proof. The G-equivariant diffeomorphism $\Phi : \mathrm{V} \to \mathrm{G} \times_{\mathrm{H}} \mathrm{E}$ induces a diffeomorphism $\bar{\Phi} : \mathrm{V}/\mathrm{G} \to (\mathrm{G} \times_{\mathrm{H}} \mathrm{E})/\mathrm{G}$. Thus, we need to show that $(\mathrm{G} \times_{\mathrm{H}} \mathrm{E})/\mathrm{G}$ is diffeomorphic to E/H. By definition, $\mathrm{G} \times_{\mathrm{H}} \mathrm{E} = (\mathrm{G} \times \mathrm{E})/\mathrm{H}$. So, $(\mathrm{G} \times_{\mathrm{H}} \mathrm{E})/\mathrm{G} = ((\mathrm{G} \times \mathrm{E})/\mathrm{H})/\mathrm{G}$. The group G acts on $\mathrm{G} \times \mathrm{E}$ by $g' \cdot (g, \xi) = (g'g, \xi)$. The group H acts on $\mathrm{G} \times \mathrm{E}$ by $h \cdot (g, \xi) = (gh^{-1}, h\xi)$. These two actions commute: $g' \cdot (h \cdot (g, \xi)) = g' \cdot (gh^{-1}, h\xi) = (g'(gh^{-1}), h\xi) = ((g'g)h^{-1}, h\xi)$. $h \cdot (g' \cdot (g, \xi)) = h \cdot (g'g, \xi) = ((g'g)h^{-1}, h\xi)$. Since the actions of G and H on $\mathrm{G} \times \mathrm{E}$ commute, the iterated quotients are diffeomorphic: $((\mathrm{G} \times \mathrm{E})/\mathrm{H})/\mathrm{G} \simeq ((\mathrm{G} \times \mathrm{E})/\mathrm{G})/\mathrm{H}$. The quotient $(\mathrm{G} \times \mathrm{E})/\mathrm{G}$ is diffeomorphic to E via the projection $\mathrm{pr}_{\mathrm{E}} : \mathrm{G} \times \mathrm{E} \to \mathrm{E}$, $\mathrm{pr}_{\mathrm{E}}(g, \xi) = \xi$. This map is G-invariant in the first factor and induces a diffeomorphism $(\mathrm{G} \times \mathrm{E})/\mathrm{G} \to \mathrm{E}$. Therefore, $((\mathrm{G} \times \mathrm{E})/\mathrm{G})/\mathrm{H} \simeq \mathrm{E}/\mathrm{H}$. Combining the diffeomorphisms, we have $\mathrm{V}/\mathrm{G} \simeq (\mathrm{G} \times_{\mathrm{H}} \mathrm{E})/\mathrm{G} \simeq \mathrm{E}/\mathrm{H}$. □

6. The Orthofold Category.

The local model E/H, where E is a Euclidean space and $\mathrm{H} \subset \mathrm{O}(\mathrm{E})$ is a compact subgroup acting orthogonally, appears fundamental for orbit spaces of compact group actions on manifolds. This suggests the definition of a class of diffeological spaces, tentatively termed *orthofolds*, characterized by being locally diffeomorphic, at every point, to such a quotient E/H, in the spirit of [PIZ13, Section "Modeling Diffeology"]. Formally:

Definition (Orthofolds). *A diffeological space* X *is called an* orthofold *if* X *is locally diffeomorphic, at each point* x*, to some quotient* E/H*, where* E *is a Euclidean vector space and* H *a subgroup of the orthogonal group* $\mathrm{O}(\mathrm{E})$*. The pair* (E,H) *is called a* local model *for the orthofold at* x*.*

This category would naturally include smooth manifolds (where H is trivial) and orbifolds (where H is finite). All orbit spaces M/G of manifolds by compact Lie group actions are natural examples of orthofolds.

Proposition (Orbifolds within Orthofolds). *A diffeological space* X *is an orthofold with a constant dimension map if and only if it is an orbifold. So, an orbifold is an orthofold of constant dimension.*

Proof. Let X be an orthofold. By definition, for any $x \in \mathrm{X}$, there is a local model diffeomorphic to E/H where E is a Euclidean space and $\mathrm{H} \subset \mathrm{O}(\mathrm{E})$ is a compact subgroup. The diffeological dimension of this local model at the point corresponding to x (the image of 0_{E}) is $\dim(\mathrm{E})$ (by the Lemma in the proof of the dimension formula).

First: If X is an n-orbifold, it is, by definition (e.g., as a diffeological space [IKZ10]), locally diffeomorphic to E/H where $\dim(\mathrm{E}) = n$ and H is a *finite* subgroup of $\mathrm{O}(\mathrm{E})$. The diffeological dimension of such a local model E/H (with H finite) is $\dim(\mathrm{E}) = n$ at every point within that model (since finite group actions on E have zero-dimensional orbits outside the origin, and the dimension formula $\dim_{[v]_{\mathrm{H}}}(\mathrm{E}/\mathrm{H}) = \dim(\mathrm{E}) - \dim(\mathrm{H}\cdot v)$ applies). Since this holds for all local models, an n-orbifold has a constant dimension map equal to n.

Next: Assume X is an orthofold with a constant dimension map, say $\dim_y(\mathrm{X}) = d$ for all $y \in \mathrm{X}$. Consider any local model E/H for X. We must have $\dim(\mathrm{E}) = d$. For any point $[v]_{\mathrm{H}} \in \mathrm{E}/\mathrm{H}$ (where $v \in \mathrm{E}$), the dimension of E/H at this point is given by the formula $\dim_{[v]_{\mathrm{H}}}(\mathrm{E}/\mathrm{H}) = \dim(\mathrm{E}) - \dim(\mathscr{O}_v)$, where $\mathscr{O}_v = \mathrm{H}\cdot v$ is the orbit of v under H within E (this formula is analogous to Proposition "Quotient Dimension-Map Formula" of article 7 applied to the action of H on E). Since the dimension map is constant and equal to $d = \dim(\mathrm{E})$, we must have $\dim(\mathrm{E}) - \dim(\mathscr{O}_v) = \dim(\mathrm{E})$ for all $v \in \mathrm{E}$. This implies $\dim(\mathscr{O}_v) = 0$ for all $v \in \mathrm{E}$. For a compact Lie group H acting orthogonally on E, if all its orbits $\mathrm{H}\cdot v$ are zero-dimensional, then H must be a discrete group [PIZ13, §1.81]. Since H is also compact, it must be a finite group. Therefore, every local model for X is of the form E/H with H finite. This means X is an orbifold. ☐

Note. It is worth noting that other classes of diffeological spaces also share the property of having a constant diffeological dimension, such as *quasifolds* [IZP21]. By definition, a quasifold is locally diffeomorphic to a quotient $\mathbf{R}^n/\Gamma$, where Γ is a countable subgroup of the affine group $\mathrm{Aff}(\mathbf{R}^n)$ such that the orbits are discrete. At a point x in such a local model, the isotropy group $\Gamma_x = \{\gamma \in \Gamma \mid \gamma(x) = x\}$ is a countable subgroup of $\mathrm{Aff}(\mathbf{R}^n)$ fixing x. The action of Γ_x on the tangent space at x (which serves as the slice model) is linear, given by the linear part of Γ_x, a countable subgroup of $\mathrm{GL}(n,\mathbf{R})$. For a quasifold to be an orthofold, its local models must be diffeomorphic to E/H where E is a Euclidean space and H is a compact subgroup of O(E). However, a countable subgroup of $\mathrm{GL}(n,\mathbf{R})$ can only be compact if it is finite. Therefore, quasifolds with infinite isotropy groups (i.e., strict quasifolds) cannot be locally diffeomorphic to quotients by compact groups, and thus strict quasifolds are not orthofolds.

7. The Diffeological Dimension Map and Formula.

The diffeological dimension map $\dim_x(\mathrm{X})$ is a fundamental intrinsic invariant defined point-wise for any diffeological space X [PIZ07, PIZ13]. For the orbit space M/G, equipped with the quotient diffeology, this map $\mathcal{O} \mapsto \dim_{\mathcal{O}}(\mathrm{M}/\mathrm{G})$ provides diffeologically invariant labels attached to each orbit $\mathcal{O} \in \mathrm{M}/\mathrm{G}$.

As the diffeological dimension is an invariant under local diffeomorphisms, it is constant on the Klein strata of M/G (i.e., on the orbits of the pseudogroup of local diffeomorphisms of M/G).

The diffeological dimension in M/G captures geometric characteristics of the singularities, reflecting how orbits deviate from being principal. The dimension of M/G at a point $\mathcal{O}$ corresponding to an orbit $\mathrm{G}\cdot x$ is given by the following formula:

Proposition (Quotient Dimension-Map Formula). *Let* $\mathrm{orbit} : \mathrm{M} \to \mathrm{M}/\mathrm{G}$ *be the projection from* M *onto the space of orbits* M/G, *equipped with the quotient diffeology. For a point* $x \in \mathrm{M}$, *let* $\mathcal{O} = \mathrm{orbit}(x)$ *be the orbit* $\mathrm{G}\cdot x$. *The diffeological dimension of* M/G *at the point* $\mathcal{O}$ *obeys the formula:*

$$\dim_{\mathcal{O}}(\mathrm{M}/\mathrm{G}) = \dim(\mathrm{M}) - \dim(\mathcal{O}). \tag{$\clubsuit$}$$

Proof. The formula $\dim_{\mathcal{O}}(\mathrm{M}/\mathrm{G}) = \dim(\mathrm{M}) - \dim(\mathcal{O})$ relies on the local structure of M/G and the dimension of the local models.

Lemma. Let E be a finite-dimensional real vector space, and let $\mathrm{H} \subset \mathrm{O}(\mathrm{E})$ be a compact subgroup of its orthogonal group. Then $\dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{E}/\mathrm{H}) = \dim(\mathrm{E})$.

Proof of the Lemma. Let $\Delta = \mathrm{E}/\mathrm{O}(\mathrm{E})$ and $\pi_{\mathrm{O}(\mathrm{E})} : \mathrm{E} \to \Delta$ be the projection. Let $\mathrm{X} = \mathrm{E}/\mathrm{H}$ and $\pi_\mathrm{H} : \mathrm{E} \to \mathrm{X}$ be the projection. Define $\mathrm{pr} : \mathrm{X} \to \Delta$ by $\mathrm{pr}([v]_\mathrm{H}) = [v]_{\mathrm{O}(\mathrm{E})}$. All three projections are subductions: $\pi_{\mathrm{O}(\mathrm{E})}$ and π_H by definition of quotient diffeology, and pr because it is induced by the identity on E and respects the equivalence relations. These maps send the origin of E to the respective origins of the quotient spaces: $\pi_\mathrm{H}(0_\mathrm{E}) = [0_\mathrm{E}]_\mathrm{H}$, $\pi_{\mathrm{O}(\mathrm{E})}(0_\mathrm{E}) = [0_\mathrm{E}]_{\mathrm{O}(\mathrm{E})}$, and $\mathrm{pr}([0_\mathrm{E}]_\mathrm{H}) = [0_\mathrm{E}]_{\mathrm{O}(\mathrm{E})}$. Since π_H and pr are subductions preserving the origins (in the sense that the origin of the domain maps to the point we are considering as the "origin" in the codomain), by properties of the diffeological dimension under subductions (see [PIZ13, §2.24, 2.26]), we have:

$$\dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{X}) \le \dim_{0_\mathrm{E}}(\mathrm{E}), \text{ and } \dim_{[0_\mathrm{E}]_{\mathrm{O}(\mathrm{E})}}(\Delta) \le \dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{X}).$$

This yields the chain:

$$\dim_{[0_\mathrm{E}]_{\mathrm{O}(\mathrm{E})}}(\Delta) \le \dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{X}) \le \dim_{0_\mathrm{E}}(\mathrm{E}).$$

The diffeological dimension of E at any point is its manifold dimension, $\dim_{0_\mathrm{E}}(\mathrm{E}) = \dim(\mathrm{E})$. From [PIZ13, Exercise 50] (with $\mathrm{E} = \mathbf{R}^n$), we know that $\dim_{[0_\mathrm{E}]_{\mathrm{O}(\mathrm{E})}}(\Delta) = \dim(\mathrm{E})$.

Thus,

$$\dim(\mathrm{E}) \leq \dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{X}) \leq \dim(\mathrm{E}).$$

This forces $\dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{X}) = \dim(\mathrm{E})$, that is, $\dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{E}/\mathrm{H}) = \dim(\mathrm{E})$. ☐

Now, let $\mathscr{O} = \mathrm{orbit}(x) \in \mathrm{M}/\mathrm{G}$ be the orbit $\mathrm{G} \cdot x$, with stabilizer $\mathrm{H} = \mathrm{Stab}_\mathrm{G}(x)$. By Proposition"Diffeological Local Model" of article 5, M/G is locally diffeomorphic around the point $\mathscr{O} \in \mathrm{M}/\mathrm{G}$ to E/H, where E is the slice at x. The point $\mathscr{O} \in \mathrm{M}/\mathrm{G}$ corresponds to $[0_\mathrm{E}]_\mathrm{H}$ in this local model. Since the diffeological dimension is a local invariant, $\dim_\mathscr{O}(\mathrm{M}/\mathrm{G}) = \dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{E}/\mathrm{H})$. By the Lemma, $\dim_{[0_\mathrm{E}]_\mathrm{H}}(\mathrm{E}/\mathrm{H}) = \dim(\mathrm{E})$. The dimension of the manifold M at x is related to the dimensions of the orbit and the slice by $\dim(\mathrm{M}) = \dim(\mathrm{G}\cdot x) + \dim(\mathrm{E})$. The dimension of the orbit $\mathrm{G}\cdot x$ is $\dim(\mathrm{G}) - \dim(\mathrm{Stab}_\mathrm{G}(x)) = \dim(\mathrm{G}) - \dim(\mathrm{H})$, but more relevant here is its dimension as a submanifold of M, which is $\dim(\mathscr{O})$. Thus, $\dim(\mathrm{M}) = \dim(\mathscr{O}) + \dim(\mathrm{E})$. Substituting $\dim(\mathrm{E}) = \dim_\mathscr{O}(\mathrm{M}/\mathrm{G})$, we get $\dim(\mathrm{M}) = \dim(\mathscr{O}) + \dim_\mathscr{O}(\mathrm{M}/\mathrm{G})$, which yields the formula: $\dim_\mathscr{O}(\mathrm{M}/\mathrm{G}) = \dim(\mathrm{M}) - \dim(\mathscr{O})$. ☐

8. Consequences of the Dimension Formula.

The dimension formula (♣) provides insight into the structure of M/G:

(a) The set of principal orbits $(\mathrm{M}/\mathrm{G})_\mathrm{pr} = \mathrm{orbit}(\mathrm{M}_\mathrm{pr})$ is the *principal Klein stratum* of M/G. It is an open dense subset of M/G (in the D-topology, which coincides with the standard quotient topology for M/G) and consists of manifold points (i.e., points whose local model is a Euclidean space). All principal orbits $\mathscr{O}_\mathrm{pr}$ have the same dimension. Thus, the dimension map is constant on $(\mathrm{M}/\mathrm{G})_\mathrm{pr}$, with value $d_\mathrm{pr} = \dim(\mathrm{M}) - \dim(\mathscr{O}_\mathrm{pr})$, which is consistent with the general property in diffeology that the dimension map is constant on Klein strata.

(b) For any singular orbit $\mathscr{O}_s$, its dimension $\dim(\mathscr{O}_s)$ is less than or equal to the dimension of a principal orbit $\dim(\mathscr{O}_\mathrm{pr})$. This is because the stabilizer of a singular point contains a conjugate of a principal stabilizer, so $\dim(\mathrm{Stab}_\mathrm{G}(s)) \geq \dim(\mathrm{Stab}_\mathrm{G}(x_\mathrm{pr}))$, which implies $\dim(\mathscr{O}_s) = \dim(\mathrm{G}) - \dim(\mathrm{Stab}_\mathrm{G}(s)) \leq \dim(\mathrm{G}) - \dim(\mathrm{Stab}_\mathrm{G}(x_\mathrm{pr})) = \dim(\mathscr{O}_\mathrm{pr})$. By formula (♣), $\dim_{\mathscr{O}_s}(\mathrm{M}/\mathrm{G}) = \dim(\mathrm{M}) - \dim(\mathscr{O}_s) \geq \dim(\mathrm{M}) - \dim(\mathscr{O}_\mathrm{pr}) = d_\mathrm{pr}$. Thus, d_pr is the minimal dimension in M/G.

(c) Equality, $\dim_{\mathscr{O}_s}(\mathrm{M}/\mathrm{G}) = d_\mathrm{pr}$, occurs for singular orbits $\mathscr{O}_s$ if and only if $\dim(\mathscr{O}_s) = \dim(\mathscr{O}_\mathrm{pr})$. Such orbits are often referred to as *exceptional orbits*. This happens when the stabilizer $\mathrm{Stab}_\mathrm{G}(s)$ has the same dimension as a principal stabilizer but is not conjugate to it.

9. Application to Toric Actions and Depth.

The diffeological dimension map provides a natural way to quantify local complexity in quotient spaces arising from group actions, sometimes aligning with or refining existing topological or geometric invariants. Consider an effective action of the n-torus T^n on a $2n$-dimensional symplectic manifold $(\mathrm{M}^{2n}, \omega)$. Such actions are central to symplectic geometry, and their orbit spaces $\mathrm{Q}_n = \mathrm{M}^{2n}/\mathrm{T}^n$ have a well-studied combinatorial and topological structure.

In many local models for these quotient spaces (e.g., near a fixed point or a point with a non-trivial stabilizer), the structure is locally diffeomorphic to a quotient of the form E/H where E is a Euclidean space and H is a compact subgroup of O(E) acting orthogonally. A standard example of such a local model is $\mathbf{C}^n/\mathrm{T}^n$, where T^n acts by $(e^{i\theta_1},\ldots,e^{i\theta_n})\cdot(z_1,\ldots,z_n)=(e^{i\theta_1}z_1,\ldots,e^{i\theta_n}z_n)$. The quotient $\mathbf{C}^n/\mathrm{T}^n$ is diffeomorphic as a diffeological space to $[0,\infty[^n$ equipped with the quotient diffeology, via the map $(z_1,\ldots,z_n)\mapsto(|z_1|^2,\ldots,|z_n|^2)$.

In the topological analysis of such spaces, a label called the *depth* is often associated with points in the quotient, see for example [KS]. For a point $t=(t_1,\ldots,t_n)\in[0,\infty[^n$ representing an orbit, its depth is typically defined as the number of coordinates t_i that are equal to zero. This corresponds to the dimension of the subtorus in T^n that stabilizes points in $\mathbf{C}^n$ projecting to t.

Let $t=(|z_1|^2,\ldots,|z_n|^2)$ be a point in $\mathbf{C}^n/\mathrm{T}^n$. According to the dimension formula (♣), the diffeological dimension at t is $\dim_t(\mathbf{C}^n/\mathrm{T}^n)=\dim(\mathbf{C}^n)-\dim(\mathcal{O}_z)$, where $\mathcal{O}_z=\mathrm{T}^n\cdot z$. The dimension of the orbit $\mathcal{O}_z$ is $\dim(\mathrm{T}^n)-\dim(\mathrm{Stab}_{\mathrm{T}^n}(z))$. The stabilizer $\mathrm{Stab}_{\mathrm{T}^n}(z)$ is isomorphic to T^k if exactly k coordinates of z are zero. Thus, $\dim(\mathrm{Stab}_{\mathrm{T}^n}(z))=k$, and $\dim(\mathcal{O}_z)=n-k$. The integer k is precisely the depth of the point t.

Substituting this into the formula:

$$\dim_t(\mathbf{C}^n/\mathrm{T}^n)=\dim(\mathbf{C}^n)-(n-k)=2n-(n-k)=n+k.$$

Thus, the depth, an important combinatorial invariant in the study of toric symplectic manifolds, is directly related to the diffeological dimension of this local model of the orbit space by the formula:

$$\dim_t(\mathbf{C}^n/\mathrm{T}^n)=n+\mathrm{depth}(t).$$

This shows how the diffeological dimension naturally incorporates and quantifies the "degree of singularity" captured by the depth in this context. For a point t in the interior of $[0,\infty[^n$ (depth $k=0$), $\dim_t(\mathbf{C}^n/\mathrm{T}^n)=n$. For a point at the origin of $[0,\infty[^n$ (depth $k=n$), $\dim_t(\mathbf{C}^n/\mathrm{T}^n)=2n=\dim(\mathbf{C}^n)$.

A final remark: It is important to distinguish this quotient diffeology on $[0,\infty[^n$ (obtained from $\mathbf{C}^n/\mathrm{T}^n$) from its standard diffeology as a manifold with corners when viewed as a subset of $\mathbf{R}^n$. For the latter, the diffeological dimension is ∞ at any point on the boundary, and n for interior points. In contrast, the quotient diffeology yields $\dim_t(\mathbf{C}^n/\mathrm{T}^n)=n+\mathrm{depth}(t)$ for all t, reflecting the structure inherited from $\mathbf{C}^n$ and the T^n-action. Rather than augmenting the topological structure of a manifold with corners with additional combinatorial data, the quotient diffeology intrinsically captures this stratification. The diffeological dimension alone recovers the depth, providing a unified and coherent geometric framework.

III. Examples

This section presents several examples of quotients of manifolds by compact group actions. They are selected to illustrate the intricate, and often surprising, relationship between the geometry of the action on M and the intrinsic Klein stratification of the quotient space M/G. Each example reveals a subtlety that challenges the simple expectation of a direct correspondence and motivates the more refined analysis presented in the subsequent sections.

10. Example $(\mathrm{S}^2 \times \mathrm{S}^2)/\,\mathrm{SO}(3)$.

This first example illustrates a baseline case where the relationship appears simple. The geometry of the action on M projects neatly onto the Klein stratification of the quotient.

We consider the group SO(3) acting diagonally on $\mathrm{M} = \mathrm{S}^2 \times \mathrm{S}^2$. The stabilizers are:

$$\mathrm{Stab}_{\mathrm{SO}(3)}(x,y) = \begin{cases} \{\mathrm{id}\} & \text{if } x \neq y \text{ and } x \neq -y \;\; \text{(principal type)}, \\ \mathrm{SO}(2,x) & \text{if } x = y \text{ or } x = -y \;\; \text{(singular type)}, \end{cases}$$

where $\mathrm{SO}(2,x)$ is the subgroup fixing x. All singular stabilizers are conjugate to SO(2).

Thus, M splits into two subsets according to the stabilizer subgroup: the principal set $\mathrm{M}_{\{\mathrm{id}\}}$ and the singular set $\mathrm{M}_{\mathrm{SO}(2)}$.

The orbit space $\mathrm{Q} = \mathrm{M}/\,\mathrm{SO}(3)$ is diffeomorphic to the interval $[-1,+1]_\pi$, where the map $\pi(x,y) = \langle x,y \rangle$ distinguishes the orbits. The Klein stratification of Q consists of two strata: the interior $\mathrm{K}_1 = \,]{-1},1[$ and the boundary $\mathrm{K}_2 = \{\pm 1\}$.

The projection of the corresponding stabilizer types aligns perfectly: $\mathrm{orbit}(\mathrm{M}_{(\{\mathrm{id}\})}) = \mathrm{K}_1$ and $\mathrm{orbit}(\mathrm{M}_{(\mathrm{SO}(2))}) = \mathrm{K}_2$. This example suggests a simple correspondence might hold in general.

11. Example $\mathrm{P}^2(\mathbf{R})/\,\mathrm{SO}(2)$.

This example shows that different types of singularities can coexist within a simple quotient space. Consider the action of $\mathrm{G} = \mathrm{SO}(2,k)$ (rotations around the unit vector k) on the real projective plane $\mathrm{M} = \mathrm{P}^2(\mathbf{R}) = \mathrm{S}^2/\{\pm 1\}$. The orbit space $\mathrm{Q} = \mathrm{M}/\mathrm{G}$ is diffeomorphic to the interval $[0,1]_\pi$, with $\pi : [x] \mapsto \langle x,k \rangle^2$ (the quotient map for the quotient diffeology).

An analysis of the stabilizers and slice representations reveals three distinct local models in the quotient:

- ∗ The interior $]0,1[$ consists of principal points, with local model $\mathbf{R}$.
- ∗ The endpoint $t = 0$ corresponds to an orbit whose stabilizer is O(1) (the rotation by π in $\mathrm{SO}(2,k)$, which acts as -1 on the line orthogonal to k), making it an **orbifold point**. Its diffeological dimension is 1.
- ∗ The endpoint $t = 1$ corresponds to a fixed point. Its local model is $\mathbf{R}^2/\mathrm{O}(2) = \Delta_2$, making it a more general **orthofold point**. Its diffeological dimension is 2.

In this case, the three stabilizer types project to three distinct Klein strata ($\{0\}$, $\{1\}$, and $]0,1[$). The key pedagogical feature is that the quotient space is a simple segment whose endpoints represent fundamentally different kinds of singularities, a distinction clearly captured by the diffeology.

12. Example $\mathrm{P}^2(\mathbf{C})/\,\mathrm{SO}(3)$.

This example immediately complicates the picture, showing that distinct strata types in M can map into the same Klein stratum in the quotient.

Consider the complex projective plane $\mathrm{M} = \mathrm{P}^2(\mathbf{C})$ with a specific action of $\mathrm{G} = \mathrm{SO}(3)$ (see [PI91]). The stabilizers that occur determine three corresponding subsets of M:

- ∗ A principal subset $\mathrm{M}_{(\mathbf{Z}_2)}$.

* A singular subset $M_{(SO(2))}$.
* Another singular subset $M_{(O(2))}$.

The groups SO(2) and O(2) are not conjugate in SO(3), so these are genuinely different stabilizer types. The orbit space $Q = M/SO(3)$ is again diffeomorphic to an interval, say $[0,1]_\pi$, with $\pi : [X,Y] \mapsto 1-4\|X\wedge Y\|^2$ the projection from $P^2(C)$ onto $[0,1]$, which induces the quotient diffeology.

These subsets project as follows:

$$\text{orbit}(M_{(\mathbf{Z}_2)}) =]0,1[\,,\ \text{orbit}(M_{(SO(2))}) = \{0\},\ \text{and orbit}(M_{(O(2))}) = \{1\}.$$

However, an analysis of the local diffeological structure reveals a surprise. The local models are:

* For $t \in]0,1[$: model $\mathbf{R}$.
* For $t = 0$: model is diffeomorphic to $\mathbf{R}^2/O(2) = \Delta_2$.
* For $t = 1$: model is also $\mathbf{R}^2/O(2) = \Delta_2$.

Since the local models at $t = 0$ and $t = 1$ are diffeomorphic, these two points belong to the *same* Klein stratum. The Klein stratification of Q consists of only two strata: $K_1 =]0,1[$ and $K_2 = \{0,1\}$.

Thus, although the stabilizers SO(2) and O(2) are distinct, their slice representations yield diffeomorphic local quotient models. This is the key phenomenon: different stabilizers can produce the same local geometry in the quotient. The intrinsic geometry of M/G therefore merges these two configurations into a single Klein stratum.

13. Example: $P^2(\mathbf{C})/U(1)$.

This final example reveals an even deeper subtlety in the relationship between a group action and its quotient: points with identical stabilizers can give rise to fundamentally different local geometries in the quotient space. This demonstrates that no classification based solely on algebraic data attached to points in the original manifold can fully capture the intrinsic structure of the quotient.

This example is from [Aud91]. Consider the action of $G = U(1)$ on $M = P^2(\mathbf{C})$ given by $\tau\cdot[z_1 : z_2 : z_3] = [z_1 : \tau z_2 : \tau^2 z_3]$. The action has three distinct stabilizers that occur: the trivial group $\{1\}$, the two-element group $\{\pm 1\}$, and the full group U(1). The fixed points of the action are $P_0 = [1:0:0]$, $P_1 = [0:1:0]$, and $P_2 = [0:0:1]$, each with stabilizer U(1).

From the perspective of the original manifold, these three points appear identical with respect to the action: they are all fixed by the entire group. One might therefore expect their images in the quotient, $\text{orbit}(P_0)$, $\text{orbit}(P_1)$, and $\text{orbit}(P_2)$, to belong to the same Klein stratum of the quotient space M/G. However, the intrinsic geometry of the quotient reveals a finer distinction.

The local structure of the quotient near each of these points is determined not by the stabilizer alone, but by the slice representation at the point. At the three fixed points, these representations are:

* At P_0: Slice action $\rho_0(\tau)\cdot(w_1, w_2) = (\tau w_1, \tau^2 w_2)$.
* At P_1: Slice action $\rho_1(\tau)\cdot(w_1, w_2) = (\bar\tau w_1, \tau w_2)$.

∗ At P_2: Slice action $\rho_2(\tau)\cdot(w_1, w_2) = (\bar\tau^2 w_1, \bar\tau w_2)$.

The representations ρ_0 and ρ_2 are equivalent, but ρ_1 is not equivalent to them. This difference has profound consequences for the intrinsic Klein stratification of the quotient:

The slice actions at P_0 and P_2 admit non-trivial finite stabilizers on subspaces of the slice. Consequently, in the quotient, the points orbit(P_0) and orbit(P_2) lie in the closure of points whose preimages have finite nontrivial stabilizers. They are adjacent to a lower-dimensional Klein stratum of the quotient.

Conversely, the slice action at P_1 is free away from the origin, meaning that near orbit(P_1), all nearby points in the quotient come from orbits with trivial stabilizer. Thus orbit(P_1) is isolated from this lower-dimensional Klein stratum.

Therefore, orbit(P_0) and orbit(P_2) lie in one Klein stratum of M/G, while orbit(P_1) lies in a completely different one. The quotient space itself, through its intrinsic Klein stratification, distinguishes between points whose preimages appeared indistinguishable when viewed only through the lens of stabilizer data.

This phenomenon—where points with identical stabilizers are separated by the projection into distinct Klein strata of the quotient—illustrates a central principle of our work: the quotient space is not merely a shadow of the action, but a new geometric object whose intrinsic structure actively organizes the original manifold. This is precisely why this decomposition — which records not just the stabilizer but also the slice representation via its connected components — is necessary to understand the quotient's intrinsic structure. Understanding this structure requires looking beyond stabilizers to the full local geometry encoded in slice representations.

These examples collectively motivate the need for a partition of M, which will be defined in the following article, that is sensitive not only to the stabilizer of a point, but also to the specific slice representation. This is precisely what this decomposition captures, and it will be defined and analyzed in the following sections.

14. Example: Orbifolds.

The case of orbifolds provides a useful classical benchmark against which to compare the more complex behaviors observed in our examples.

An n-orbifold is a diffeological space locally diffeomorphic to $\mathbf{R}^n/\Gamma$ for some finite subgroup $\Gamma \subset \mathrm{GL}(n, \mathbf{R})$. The conjugacy class of Γ is the structure group at that point.

A key result from [IKZ10, Lemma 26] states that two points in an orbifold belong to the same Klein stratum if and only if their structure groups are conjugate.

When an orbifold arises as a quotient M/G of a manifold by a finite group G, the structure group at orbit(x) is precisely the stabilizer $\mathrm{Stab}_G(x)$. In this situation, the Klein stratification of the quotient M/G corresponds exactly to the partition of M by stabilizer conjugacy classes projected to the quotient. Points whose preimages have conjugate stabilizers lie in the same Klein stratum, and distinct conjugacy classes give rise to distinct strata. This direct correspondence holds for finite group actions but, as shown in the previous examples, fails when the group has non-trivial identity components.

This direct correspondence between the data attached to points in M and the intrinsic Klein stratification of M/G holds in the finite group setting. It serves as a crucial benchmark, highlighting that the more complex behaviors observed in our other examples—where points with identical stabilizers map to distinct Klein strata, and points with non-conjugate stabilizers map to the same stratum—are characteristic of actions involving stabilizers with non-trivial identity components (such as SO(2) and U(1)).

In those cases, the intrinsic geometry of the quotient encodes information beyond stabilizer data alone, captured by the slice representation. This motivates the need for a finer partition of M that is sensitive to this additional local structure.

IV. The Isostabilizer Decomposition

The examples in the previous section demonstrate the need for a partition of M that is sensitive not only to the stabilizer subgroup itself but also to the equivalence class of its action on the normal slice. This section formally defines this partition, the isostabilizer decomposition, which provides the precise relationship between the geometry of the action and the intrinsic structure of the quotient.

Let G be a compact Lie group acting smoothly on a connected smooth manifold M.

15. Orbits and Stabilizers.

For any point $x \in \mathrm{M}$, its *orbit* is the set $\mathrm{G} \cdot x = \{g \cdot x \mid g \in \mathrm{G}\}$. The *stabilizer* (or isotropy group) of x is the subgroup $\mathrm{Stab}_{\mathrm{G}}(x) = \{g \in \mathrm{G} \mid g \cdot x = x\}$. The stabilizer is a closed subgroup of G, and all points on the same orbit have conjugate stabilizers: $\mathrm{Stab}_{\mathrm{G}}(g \cdot x) = g\,\mathrm{Stab}_{\mathrm{G}}(x) g^{-1}$.

16. Isostabilizer Submanifolds.

For each closed subgroup $\mathrm{K} \subseteq \mathrm{G}$, we define the *isostabilizer set* $\mathrm{M}_{\mathrm{K}} = \{x \in \mathrm{M} \mid \mathrm{Stab}_{\mathrm{G}}(x) = \mathrm{K}\}$. The collection of non-empty sets $\{\mathrm{M}_{\mathrm{K}}\}$ for all closed subgroups $\mathrm{K} \subseteq \mathrm{G}$ forms a partition of M. Each of these sets is, in fact, a submanifold.

Theorem (Isostabilizer Sets are Submanifolds). *For any closed subgroup* $\mathrm{K} \subseteq \mathrm{G}$*, the set* M_{K} *is a smooth submanifold of* M *(possibly empty).*

Proof. By the Differentiable Slice Theorem [Pal60, Bre72], for any $x \in \mathrm{M}_{\mathrm{K}}$, a G-invariant open neighborhood U of the orbit $\mathrm{G}\cdot x$ is G-equivariantly diffeomorphic to an associated bundle $\mathrm{G} \times_{\mathrm{K}} \mathrm{B}$, where B is a K-invariant open ball in the normal space N_x on which K acts linearly. The subset $\mathrm{M}_{\mathrm{K}} \cap \mathrm{U}$ corresponds under this diffeomorphism to the points $[g, v]_{\mathrm{K}} \in \mathrm{G} \times_{\mathrm{K}} \mathrm{B}$ with stabilizer K. This condition holds if and only if $g \in \mathrm{N(K)}$ (the normalizer of K) and $v \in \mathrm{B}^{\mathrm{K}}$ (the set of K-fixed points in B).

Thus, $\mathrm{M}_{\mathrm{K}} \cap \mathrm{U}$ is diffeomorphic to the bundle $\mathrm{N(K)} \times_{\mathrm{K}} \mathrm{B}^{\mathrm{K}}$, which is trivial and hence diffeomorphic to the product $(\mathrm{N(K)}/\mathrm{K}) \times \mathrm{B}^{\mathrm{K}}$. The quotient $\mathrm{N(K)}/\mathrm{K}$ is a manifold. Since the action of K on N_x is linear, its fixed-point set $(\mathrm{N}_x)^{\mathrm{K}}$ is a linear subspace. The set $\mathrm{B}^{\mathrm{K}} = \mathrm{B} \cap (\mathrm{N}_x)^{\mathrm{K}}$ is an open subset of this subspace and is therefore a smooth manifold. The product of two manifolds is a manifold, proving that M_{K} is locally a manifold, and thus is a manifold itself. ◻

Remark. While this is a standard result, we include the proof to highlight the local product structure $(\mathrm{N}(\mathrm{K})/\mathrm{K}) \times \mathrm{B}^{\mathrm{K}}$. This structure is the foundational brick for the fine analysis of the quotient diffeology.

17. The Isostabilizer Decomposition.

The local structure of the quotient space M/G at a point $\mathrm{orbit}(x)$ is determined by the action of the stabilizer $\mathrm{Stab}_{\mathrm{G}}(x)$ on the slice at x. As shown in Example 13, points with the same stabilizer K can still produce different local structures in the quotient if their slice representations are not equivalent. This motivates looking at the sets where the slice representation remains constant.

Proposition (Slice Equivalence on Components). *Let* C *be a connected component of an isostabilizer submanifold* M_{K}*. For any two points* $x, x' \in \mathrm{C}$*, the slice representations* $\rho_x : \mathrm{K} \to \mathrm{O}(\mathrm{N}_x)$ *and* $\rho_{x'} : \mathrm{K} \to \mathrm{O}(\mathrm{N}_{x'})$ *are equivalent as representations of* K*.*

Proof. Let C be a connected component of M_{K}. We can equip M with a G-invariant Riemannian metric. The normal spaces N_x to the orbits $\mathrm{G} \cdot x$ for $x \in \mathrm{C}$ form a smooth vector bundle $\mathscr{N} \to \mathrm{C}$. The slice representation at x is the action of K on the fiber N_x. Since C is connected, we can connect any two points x, x' by a path. The continuous dependence of the representation along this path, combined with the fact that the space of equivalence classes of representations of a compact group on a fixed-dimensional vector space is discrete, implies that the representations ρ_x and $\rho_{x'}$ must be equivalent (see, e.g., [BT85, Fol95]). □

This proposition shows that the connected components of the isostabilizer submanifolds are the fundamental "atoms" of the action, on which the local data (stabilizer and slice representation) is constant. This justifies the following central definition.

Definition (Isostabilizer Decomposition). *The* isostabilizer decomposition *of the* G-*manifold* M *is the partition of* M *into the connected components of the isostabilizer submanifolds* $\{\mathrm{M}_{\mathrm{K}}\}$*. We denote the set of these components by* $\mathrm{Iso}(\mathrm{M}, \mathrm{G})$.

This decomposition is finer than a partition based solely on stabilizers. As we will show in the next section, it is precisely this decomposition of M that maps coherently to the Klein stratification of the quotient space M/G.

V. The Main Correspondence Theorem

The previous sections have established the necessary tools: the intrinsic Klein stratification of the quotient space M/G and the refined isostabilizer decomposition of the manifold M. We are now in a position to state and prove the precise relationship between them. We will show that the isostabilizer decomposition of M is exactly the structure that maps coherently to the Klein stratification of M/G.

18. The Induced Map on Decompositions.

Recall that $\mathrm{Iso}(\mathrm{M}, \mathrm{G})$ is the set of connected components of the isostabilizer submanifolds M_{K}, and $\mathrm{K}\ell(\mathrm{M}/\mathrm{G})$ is the set of Klein strata of the quotient space. By Proposition "Slice Equivalence on Components" of article 17, for any component $\mathrm{C} \in \mathrm{Iso}(\mathrm{M}, \mathrm{G})$, the stabilizer K is constant and the slice representation is fixed up to equivalence. This implies that the local model for the quotient, E_x/K, is of a constant diffeomorphic type for all points $x \in C$.

Since the Klein strata are, by definition, the sets of points with diffeomorphic local models, the image of any component C under the orbit projection must lie entirely within a single Klein stratum. This observation leads to our main theorem.

Theorem (The Correspondence Theorem). *The canonical projection* $\mathrm{orbit} : M \to M/G$ *induces a well-defined, surjective map*

$$\mathrm{orbit}_* : \mathrm{Iso}(M, G) \to K\ell(M/G)$$

from the isostabilizer decomposition of M *to the Klein stratification of* M/G.

Proof. **Well-definedness:** Let $C \in \mathrm{Iso}(M, G)$. By Proposition "Slice Equivalence on Components", for any two points $x_1, x_2 \in C$, their stabilizers are the same group K, and their slice representations are equivalent. The local model of the quotient M/G at a point $\mathrm{orbit}(x)$ is determined by the equivalence class of the slice representation of $\mathrm{Stab}_G(x)$. Since this is constant for all $x \in C$, the points $\mathrm{orbit}(x_1)$ and $\mathrm{orbit}(x_2)$ have diffeomorphic local models. Therefore, they belong to the same Klein stratum. This shows that the entire set orbit(C) is contained within a single Klein stratum, so the map orbit_* is well-defined.

Surjectivity: Let $\mathscr{K} \in K\ell(M/G)$ be an arbitrary Klein stratum. By definition, there must exist at least one point $\mathscr{O} \in \mathscr{K}$. This point is an orbit, so $\mathscr{O} = \mathrm{orbit}(x)$ for some $x \in M$. This point x must belong to some isostabilizer submanifold M_K, and therefore to one of its connected components, say $C \in \mathrm{Iso}(M, G)$. By construction, $\mathrm{orbit}_*(C)$ is the Klein stratum containing $\mathrm{orbit}(x)$, which is $\mathscr{K}$. Thus, every Klein stratum is in the image of orbit_*. □

19. The Inverse Klein Stratification.

The Correspondence Theorem allows us to define a new, canonical stratification on M induced conversely by the quotient's geometry.

Let $\{K\ell_i\}$ be the Klein stratification of M/G. We define the **Inverse Klein Stratification** of M to be the partition $\{S_i\}$ where each set $S_i = \mathrm{orbit}^{-1}(K\ell_i)$ is the preimage of a Klein stratum.

Proposition (The Inverse Klein is a Stratification). *The Inverse Klein Stratification* $\{S_i\}$ *is a true stratification of the manifold* M.

Proof. Let $K\ell_j \subset \overline{K\ell_i}$ in M/G. We must show that $S_j \subset \overline{S_i}$. Let $y \in S_j$. For any open neighborhood W of y, orbit(W) is an open neighborhood of $\mathrm{orbit}(y)$. Since $\mathrm{orbit}(y) \in \overline{K\ell_i}$, orbit(W) must intersect $K\ell_i$. This implies there exists an $x \in W$ such that $\mathrm{orbit}(x) \in K\ell_i$, meaning $x \in S_i$. Thus, any neighborhood of y intersects S_i, so $y \in \overline{S_i}$. □

Our key examples illustrate how the Inverse Klein Stratification relates to the stabilizer data on M — note that this stratification is defined purely by pullback from the quotient's intrinsic geometry, independent of any particular notion of "orbit type" *a priori* classification.

- In Example 12, points with different stabilizers (SO(2) and O(2)) map to the same Klein stratum, showing that the Inverse Klein Stratification can be coarser

than a partition based solely on stabilizers (see §21 for a detailed analysis of this phenomenon).
- In Example 13, points with identical stabilizers (U(1)) map to distinct Klein strata, showing that the Inverse Klein Stratification can be finer than such a partition (see §21 for the interpretation in terms of slice representations).

This clarifies the hierarchy of partitions on M: the isostabilizer decomposition $\{C_\alpha\}$ is the finest of the three, refining both a partition by stabilizers and the new Inverse Klein Stratification.

20. Characterization of Singularity Types in M/G.

The Correspondence Theorem allows us to classify the geometric nature of the points in the quotient space by examining the stabilizers of the corresponding points in the manifold.

Proposition (Geometric Nature of Quotient Points). *Let* $\mathcal{O} = \mathrm{orbit}(x) \in M/G$ *be a point in the orbit space, with stabilizer* $K = \mathrm{Stab}_G(x)$.

- *If* x *is a principal point, then* $\mathcal{O}$ *is a* ***manifold point*** *(i.e., its local model is a Euclidean space).*
- *If* x *is an exceptional singular point (i.e.,* $\dim(K) = \dim(K_{pr})$ *but* K *is not conjugate to* K_{pr}*), then* $\mathcal{O}$ *is a non-trivial* ***orbifold point*** *(i.e., its local model is* E/Γ *where* Γ *is a non-trivial finite group).*
- *If* x *is a non-exceptional singular point (i.e.,* $\dim(K) > \dim(K_{pr})$*), then* $\mathcal{O}$ *is a* ***general orthofold point*** *(i.e., its local model is* E/K *where the effective action of* K *is by a non-discrete compact Lie group).*

Proof. The local model at $\mathcal{O}$ is determined by the effective action of K on the slice E. *Principal:* The effective action of a principal stabilizer on its slice is trivial [Bre72, IV, 3.2(iii)]. The local model is $E/\{\mathbf{1}\} \simeq E$. *Exceptional:* The condition $\dim(K) = \dim(K_{pr})$ implies the effective action on the slice is by the finite group $\Gamma = K/K^0$. Since the point is singular, this action is non-trivial. The local model is E/Γ, which is an orbifold. *Non-exceptional Singular:* The condition $\dim(K) > \dim(K_{pr})$ implies the effective action on the slice has a non-trivial identity component. The local model is therefore an orthofold but not an orbifold. □

21. The Meaning of Non-Injectivity.

The Correspondence Theorem establishes a surjective map $\mathrm{orbit}_* : \mathrm{Iso}(M,G) \to \mathrm{K}\ell(M/G)$. As our examples in Section III conclusively demonstrate, this map is generally not injective. This non-injectivity is not a flaw in the analysis, but rather a deep insight into the nature of diffeological quotients.

Non-injectivity occurs when distinct components $C_1, C_2 \in \mathrm{Iso}(M,G)$ map to the same Klein stratum. This happens if and only if their corresponding local models are diffeomorphic, even if the components themselves are profoundly different from the perspective of the action on M. We saw two primary ways this can happen:

(1) **Different Stabilizers, Same Local Model:** In Example 12, components of $M_{SO(2)}$ and $M_{O(2)}$ mapped to the same Klein stratum. The stabilizers SO(2) and O(2)

are not even conjugate, but their actions on their respective slices produce diffeomorphic quotients.

(2) **Same Stabilizer, Different Local Models:** In Example 13, different connected components of the single set $M_{U(1)}$ mapped to *different* Klein strata. This happened because the slice representations at these fixed points were not equivalent: they do not define local diffeomorphic quotient.

This leads to a fundamental conclusion: the intrinsic smooth identity of the quotient space M/G is determined entirely by the set of local models $\{E/H\}$ that appear. The Klein stratification captures this intrinsic identity. The specific details of the action on M—such as the conjugacy classes of stabilizers or the global structure of the isostabilizer sets—can be abstracted away in the process of taking the quotient. The quotient space only remembers the local diffeomorphic types of its singularities, the local geometry.

VI. Concluding Remarks

In this paper, we have clarified the precise relationship between the smooth action of a compact Lie group G on a manifold M and the intrinsic smooth structure of the resulting orbit space M/G. By employing the tools of diffeology, we have provided a complete picture of how the local geometry of the action translates into the local geometry of the quotient.

Our investigation yielded three main results. First, we identified the partition on the manifold M that governs this correspondence: the isostabilizer decomposition, whose elements are the connected components of submanifolds where the stabilizer is constant. We showed that on each such component, the slice representation is fixed up to equivalence, making these components the fundamental "atoms" of the group action.

Second, we established the *Correspondence Theorem*, which states that the canonical projection induces a surjective map from this isostabilizer decomposition of M to the Klein stratification of M/G. This theorem provides the definitive dictionary between the structure of the action and the intrinsic singularity types of the quotient.

Third, and perhaps most profoundly, we explored the meaning of this map's *non-injectivity*. Our examples demonstrate that distinct structural elements on M can map to diffeologically identical points in the quotient. This reveals a fundamental principle: the quotient space M/G possesses an intrinsic geometric identity that can be independent of the specific details of its construction. It distills the global structure of the action on M, remembering only the local quotient geometry defined by the local diffeomorphisms, which in turn captures the diffeomorphic types of its singularities.

This perspective gives rise to the concept of an *orthofold*, a diffeological space that is locally diffeomorphic to a quotient E/H of a Euclidean space by a compact orthogonal group action. Our analysis shows that orbit spaces of compact group actions are the canonical examples of orthofolds. This class of spaces naturally contains both smooth manifolds (H is trivial) and orbifolds (H is finite) as special cases. Our characterization of the singularity types in Proposition "Geometric Nature of Quotient Points" of article 20 provides a clear hierarchy within this class: principal orbits correspond to manifold points, exceptional orbits to orbifold points, and other singular orbits to general orthofold points. The study of orthofolds as a distinct and natural class of geometric objects, motivated by the structure of orbit spaces, presents a fruitful area for future investigation. Furthermore, the set $\mathrm{Iso}(\mathrm{M},\mathrm{G})$ itself can be endowed with a quotient diffeology, turning the map orbit_* into a subduction and opening new avenues for studying the global structure of these decompositions.

The power of the diffeological framework is not limited to such quotients. As illustrated by the space of geodesics on a torus, $\mathrm{Geod}(\mathrm{T}^2)$ [PIZ25, §72],[1] diffeology provides essential tools for analyzing the intrinsic smooth structure of a wide range of singular spaces (not limited to quotients of manifolds by compact Lie groups) where traditional differential geometry may lack a suitable global language and tools.

These results underscore the utility of diffeology in providing a consistent and nuanced framework for the study of singular spaces. By capturing the complete smooth structure that remains after reduction, the diffeological approach offers a richer understanding of a quotient's geometry than its topology alone. The study of orthofolds as a distinct and natural class of geometric objects, motivated by the structure of orbit spaces, presents a fruitful area for future investigation.

Finally, while this paper focuses on the actions of compact Lie groups, the local nature of our arguments — relying primarily on the Slice Theorem — suggests that this framework could naturally extend to proper actions of non-compact Lie groups. We leave the exploration of this generalization, and the corresponding global topological considerations, for future work.

Acknowledgement of AI Assistance. During the preparation of this work, the authors acknowledge the linguistic and organizational assistance provided by the AI assistants Gemini (Google) and DeepSeek (深度求索). The authors take full responsibility for the content of the published article.

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